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A BRIEF GUIDE TO CALCULUS II

THE UNIVERSITY OF MINNESOTA

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Integration Techniques

Throughout this guide, we will present results without proof. If you would like a proof of the techniques and theorems that we use, please see "Calculus *Early Transcendentals*", volume I, 7E by James Stewart. To begin this guide, we will review basic integration formula that every calc II student should know, and then move to new techniques.

Basic Integrals

The integrals below are essential formulas the should be memorized. If you struggle with a few of them, please practice until you have committed them to memory. Also note that the constants have been left out of the table below for convenience.

$$\int x^n \, dx = \frac{x^{n+1}}{n+1}, \;\; n \neq -1$$

$$\int \frac{1}{x} \, dx = \ln|x|$$

$$\int e^x \, dx = e^x$$

$$\int a^x \, dx = \frac{1}{\ln a} a^x$$

$$\int \sin x \, dx = -\cos x$$

$$\int \cos x \, dx = \sin x$$

$$\int \sec^2 x \, dx = \tan x$$

$$\int \csc^2 x \, dx = -\cot x$$

$$\int \sec x \, \tan x \, dx = \sec x$$

$$\int \csc x \, \cot x \, dx = -\csc x$$

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \cdot \tan^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1}\left(\frac{x}{a}\right)$$

Along with these integral formulas, you should also be comfortable with the technique of ***u*-substitution**. The next two examples illustrate how u-substitution works with respect to changing bounds of integration.

Example 1.1. *Evaluate the following integral*

$$\int_0^{\pi/2} \frac{\sin x}{1+\cos x}\, dx$$

Solution: Let $u = \cos x$*, then* $du = -\sin x$*. Now we may rewrite the integral above as:*

$$-\int_1^0 \frac{1}{1+u^2}\, du$$

Notice that limits of integration have changed from 0 *and* $\pi/2$ *to* 1 *and* 0 *respectively. This is due to the fact that when the function* $u(x) = \cos(x)$ *is evaluated at* 0 *and* $\pi/2$*, we get*

This example should be straight forward, but if you feel you need more practice,choose a few extra problems to work.

$$u(0) = 1 \qquad u(\pi/2) = 0$$

Therefore our integral becomes:

$$\begin{aligned}
&\int_0^1 \frac{1}{1+u^2}\, du \\
&= \arctan(u)\Big|_0^1 \\
&= \arctan(1) - \arctan(0) \\
&= \boxed{\pi/4}
\end{aligned}$$

Integration By Parts

The first new technique of calculus II tat we will introduce is **Integration by Parts**. This technique will give a possible way to integrate products of functions. Without further ado, the formulas for integration by parts is

$$\boxed{\int u\, dv = uv - \int v\, du}$$

$$\boxed{\int_a^b u\, dv = uv\Big|_a^b - \int_a^b v\, du}$$

This equation is often remembered by the saying "the integral of u dv is uv minus the integral of v du". We will now consider three examples of integration by parts that will illuminate the possible ways to use this formula.

Example 1.2. *Evaluate* $\int_1^2 \ln x \, dx$.

As you read through this guide, note that these examples are a minimal set of problems to study from, and it will do little good to only read these problems. You should choose problems to study in each section. It is often said that "Math is not a spectator sport", this means you must do the work by yourself on your own time.

Solution: In this example, it does not seem that we have any reasonable choices for u and dv. When this occurs, it is often useful to choose dv to be dx. This leaves us with

$$\begin{aligned} u &= \ln x & dv &= dx \\ du &= \frac{1}{x} dx & v &= x \end{aligned}$$

Therefore when we use our formula for integration by parts we get:

$$\begin{aligned} \int_1^2 \ln x \, dx &= x \ln x \Big|_1^2 - \int x \frac{dx}{x} \\ &= x \ln x \Big|_1^2 - \int_1^2 dx \\ &= x \ln x \Big|_1^2 - x \Big|_1^2 \\ &= 2 \ln 2 - 0 - 2 + 1 \\ &= \boxed{2 \ln 2 - 1} \end{aligned}$$

Example 1.3. *Evaluate the integral* $\int e^\theta \sin 2\theta \, d\theta$.

Solution: In this example, the key idea is to try integration by parts until you get back to the original integral you wish to evaluate. Following this path, we have

$$\begin{aligned} u_1 &= \sin 2\theta & dv_1 &= e^\theta \\ du_1 &= 2 \cos 2\theta & v_1 &= e^\theta \end{aligned}$$

Under this choice of u and dv, the integral above becomes

$$\int e^\theta \sin 2\theta \, d\theta = \sin 2\theta \, e^\theta - 2 \int \cos 2\theta \, e^\theta \, d\theta$$

Then if we use integration by parts again with

$$\begin{aligned} u_2 &= \cos 2\theta & dv_2 &= e^\theta \\ du_2 &= -2 \sin 2\theta & v_2 &= e^\theta, \end{aligned}$$

we get:

$$\int e^{\theta} \sin 2\theta \, d\theta = \sin 2\theta \, e^{\theta} - 4 \cos 2\theta \, e^{\theta} - 4 \int \sin 2\theta \, d\theta$$

Now if we subtract the far right integral from each side, we get

$$5 \int e^{\theta} \sin 2\theta \, d\theta = \sin 2\theta \, e^{\theta} - 4 \cos 2\theta \, e^{\theta}$$

Hence the answer we get is

$$\int e^{\theta} \sin 2\theta \, d\theta = \boxed{\frac{\sin 2\theta \, e^{\theta} - 4 \cos 2\theta \, e^{\theta}}{5} + C}$$

This final example of this section will show how to use integration by parts multiple times. A type of integral that often arises in applications is the integration of a function of the form $x^n g(x)$, where $g(x)$ can be integrated n times. When faced with a situation like this, we may use a variation of integration by parts known as **Tabular Integration**.

Example 1.4. *Evaluate the integral* $\int x^2 e^{2x} \, dx$

Try to understand why this works!

Solution: To use tabular integration in solving this problem, we will make a table in which we place the derivatives of x^2 in the right column and the integrals of e^{2x} in the left column. After constructing this table, we will add these terms as depicted in the following table.

D		I
x^2	$+$	e^{2x}
$2x$	$-$	$\frac{1}{2}e^{2x}$
2	$+$	$\frac{1}{4}e^{2x}$
0		$\frac{1}{8}e^{2x}$
$=$		$\frac{1}{2}x^2e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x}$

Therefore

$$\int x^2 \, e^{2x} \, dx = \boxed{\frac{1}{2}x^2e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + C}$$

Trigonometric Identities

In this section we will focus on the trigonometric identities that will help in simplifying certain types of integrals. Before we begin, it will be important to introduce the identities that we will be using. These identities will be essential to doing the problems and therefore must be memorized. This will help you in exams when you do not have time to derive such formulas. The necessary identities are

Pythagorean Identities

$$\sin^2 x + \cos^2 x = 1$$
$$\sec^2 x = \tan^2 + 1$$
$$\csc^2 x = \cot^2 + 1$$

Half Angle Formulas

$$\sin^2 x = \frac{1}{2}(1 - \cos 2\theta) \qquad \cos^2 x = \frac{1}{2}(1 + \cos 2\theta)$$

Double Angle Formula

$$\sin 2x = 2 \sin x \cos x$$

These identities are far a from complete, but they will suffice for most of the problem that we will encounter. To illustrate this, we will present the common strategies for integrating functions of the form $f(x) = \sin^n x \cos^n x$.

Cosine is odd

If the power of cosine is odd, then we will use the Pythagorean identity to convert all but one of the cosines to $1 - \sin^2 x$ raise to some power. This remaining cosine will serves as du in the substitution $u = \sin x$. This will give the following where $n = 2k + 1$.

$$\begin{aligned}
&\int \sin^m x \cos^n x \, dx = \\
&= \int \sin^m x (\cos^2 x)^k \cos x \, dx \\
&= \int \sin^m x (1 - \sin^2 x)^k \cos x \, dx \\
&= \int u^m (1 - u^2)^k \, du
\end{aligned}$$

Sine is odd

If the power of sine is odd, then we will use the Pythagorean identity to convert all but one of the sines to $1 - \cos^2 x$ raise to some power. This remaining sine will serves as du in the substitution $u = \cos x$. This will give the following where $m = 2k + 1$

$$\begin{aligned}
&\int \sin^m x \cos^n x \, dx \\
&= \int (\sin^2 x)^k \cos^n x \sin x \, dx \\
&= \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx \\
&= -\int (1 - u^2)^k u^n \, du
\end{aligned}$$

Both Sine and Cosine are even

If the powers of sine and cosine are both even, we use the half angle or double angle formulas to reduce powers until we are left with a simpler function that can be integrated.

We will now use these ideas to work through a few examples.

We will see that these strategies can also be applied to powers of secant and tangent.

Example 1.5. *Evaluate* $\int \sin^3 \theta \cos^4 \theta \, d\theta$

Solution: We first notice that the power of sine is odd, therefore we will use the identity $\sin^2 x = 1 - \cos^2 x$.

$$\begin{aligned}
&\int \sin^3 \theta \cos^4 \theta \, d\theta \\
&= \int (1 - \cos^2 \theta) \cos^4 \theta \sin \theta \, d\theta \\
&= -\int (1 - u^2) u^4 \, du \\
&= -\int (u^4 - u^6) \, du \\
&= -\frac{u^5}{5} - \frac{u^7}{7} \\
&= \boxed{\frac{1}{7} \cos^7 \theta - \frac{1}{5} \cos^5 \theta + C}
\end{aligned}$$

Example 1.6. *Evaluate* $\int \cos^4 2x\, dx$.

Solution: To evaluate this integral, we will have to use the half angle formula to reduce powers of cosine. This will give

$$\int \cos^4 2x\, dx$$

$$= \frac{1}{4}\int \left(1+\cos 2x\right)^2 dx$$

$$= \frac{1}{4}\int \left(1+2\cos 2x+\cos^2 2x\right) dx$$

$$= \frac{x}{4}+\frac{\sin 2x}{4}+\frac{1}{4}\int \cos^2 2x\, dx$$

$$= \frac{x}{4}+\frac{\sin 2x}{4}+\frac{1}{8}\int 1+\cos 4x\, dx$$

$$= \frac{x}{4}+\frac{\sin 2x}{4}+\frac{x}{8}+\frac{\sin 4x}{32}$$

$$= \boxed{\frac{3x}{8}+\frac{\sin 2x}{4}+\frac{\sin 4x}{32}+C}$$

Example 1.7. *Evaluate* $\int \sin^2 x \cos^2 x\, dx$

Solution: To evaluate this integral, notice that the function we are integrating can be rewritten as $(\sin x \cos x)^2$*. This can be rewritten using the double angle formula to become* $(\sin 2x)^2/4$*. Now we can use the half angle formula as in the last example to rewrite this. Therefore the integral can be evaluated as.*

$$\int \sin^2 x \cos^2 x\, dx$$
$$= \frac{1}{4}\int (\sin 2x)^2\, dx$$
$$= \frac{1}{8}\int (1-\cos 4x)\, dx$$
$$= \boxed{\frac{x}{8}-\frac{\sin 4x}{32}+C}$$

In this section, we will conclude by deriving a formula for integrating secant.

Example 1.8. *Evaluate* $\int \sec x \, dx$.

Solution: For this integral, we will use a specific trick that shows how multiplication by a special term can often help in simplifying a problem. The term that we will multiply this integral by is

$$\frac{\sec x + \tan x}{\sec x + \tan x}$$

Therefore we have

$$\int \sec x \, dx$$
$$= \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} \, dx$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$$

Now we let $u = \sec x + \tan x$, *and* $du = \sec^2 x + \sec x \tan x$. *Thus we get*

$$\int \frac{1}{u} \, du$$

$$ln|u| + C$$

$$\boxed{= \ln|\sec x + \tan x| + C}$$

Please note that the examples that were provided in this section do not show all of the possible difficulties that can arise from these types of integrals.

Trigonometric Substitution

In this section we will explore a type of integral substitution known as an **inverse substitution**. When we perform u-substitution, we have a way to determine u from x. That is, we may write u as function $u(x)$. An inverse substitution instead will give a way to determine x from u. That is, we will be able to write x as a function $x(u)$. In this section, we will consider a specific set of inverse substitutions involving trigonometric functions.

When we perform an inverse substitution we must make sure that such a substitution is one-to-one.

Table of Substitutions

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\sec^2\theta - 1 = \tan^2\theta$

As commented earlier, these substitutions are only valid if the substitution is one-to-one. Since these trigonometric functions are not one to one, we must restrict θ to be within a valid domain. The table below gives the valid domains for these substitutions.

Substitution	Interval
$x = a\sin\theta$	$\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$x = a\tan\theta$	$\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$x = a\sec\theta$	$\theta \in \left[0, \frac{\pi}{2}\right]$ or $\theta \in \left[\pi, \frac{3\pi}{2}\right]$

Before working some example, we remark that the reason these substitutions are called "inverse" substitutions is because in order to rewrite θ in terms of x we must invert any final answer we get. To see this in action, let's work some examples.

Example 1.9. *Evaluate* $\int x^3\sqrt{1-x^2}\,dx$

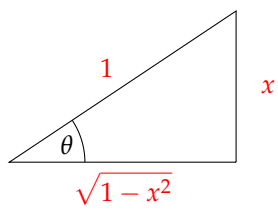
Solution: For this example, we will let $x = \sin\theta$*, then* $dx = \cos\theta$*. Then*

$$\begin{aligned}
&\int x^3\sqrt{1-x^2}\,dx \\
&= \int \sin^3\theta\sqrt{1-\sin^2\theta}\cos\theta\,d\theta \\
&= \int \sin^3 x\cos^2 x\,dx \\
&= \int \sin^2 x\cos^2 x\sin x\,dx \\
&= \int (1-\cos^2 x)\cos^2 x\sin x\,dx
\end{aligned}$$

Now if we use a u-substitution with $u = \cos\theta$*, we have the following:*

$$\int u^4 - u^2\,du$$
$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$
$$= \frac{\cos^5\theta}{5} + \frac{\cos^3\theta}{3} + C$$

Finally we must invert this equation to get back to x. In this case we use right triangle trigonometry to determine what $\cos\theta$ is in terms of x. Since $\sin\theta = \frac{x}{1}$, we will represents the transformation with the following triangle.



Therefore $\cos\theta = \sqrt{1-x^2}$, and our answer becomes

$$\boxed{\frac{(1-x^2)^{5/2}}{5} - \frac{(1-x^2)^{3/2}}{3}C}$$

Example 1.10. *Evaluate* $\int \frac{1}{\sqrt{4x^2+1}}\,dx$.

Solution: For this example, we will proceed as above and let $x = \frac{1}{2}\tan\theta$. Then $dx = \frac{1}{2}\sec^2\theta$ our problem becomes

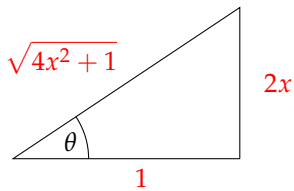
$$\int \frac{1}{\sqrt{4x^2+1}}\,dx$$

$$= \frac{1}{2}\int \frac{\sec^2\theta}{\sec\theta}\,d\theta$$

$$= \frac{1}{2}\int \sec\theta\,d\theta$$

$$= \frac{1}{2}\ln|\sec\theta + \tan\theta| + C$$

Again, we will use right triangle trigonometry to invert this expression. The diagram below will help us do so.



Hence $\tan\theta = 2x$. *Then our answer becomes*

$$\boxed{\frac{1}{2}\ln|\sqrt{4x^2+1}+2x|+C}$$

Our final example will be to show how the limits of integration change when using a trigonometric substitution.

Example 1.11. *Evaluate* $\displaystyle\int_{\sqrt{3}}^{2}\frac{\sqrt{x^2-3}}{x}\,dx$.

Solution: For this problem we will use the trigonometric substitution

$$x = \sqrt{3}\sec\theta \text{ and } dx = \sqrt{\sec\theta\tan\theta\,d\theta}$$

The key point in this example is in converting the limits of integration. To do this, we consider x *to be a function of* θ *and set* $\sqrt{3} = \sqrt{3}\sec\theta$ *and* $2 = \sqrt{3}\sec\theta$. *These equations become* $\sec\theta = 1$ *and* $\sec\theta = 2/\sqrt{3}$, *and they have the solutions* $\theta = 0$ *and* $\theta = \pi/6$. *Therefore the integral becomes*

$$\int_{\sqrt{3}}^{2}\frac{\sqrt{x^2-3}}{x}\,dx$$

$$\int_{0}^{\pi/6}\frac{(\sqrt{3}\tan\theta)(\sqrt{3}\sec\theta\tan\theta)}{\sqrt{3}\sec\theta}\,d\theta$$

$$= \int_{0}^{\pi/6}\sqrt{3}\tan^2\theta\,d\theta$$

$$\sqrt{3}\int_{0}^{\pi/6}(\sec^2\theta-1)\,d\theta$$

$$\sqrt{3}\Big[\tan\theta-\theta\Big]_{0}^{\pi/6}$$

$$= \boxed{1-\frac{\sqrt{3}\pi}{6}}$$

Partial Fraction

Another great technique that we can use for simplifying integrals is the method of partial fractions. The following few steps give a general method for decomposing a rational function $P(x)/Q(x)$.

1. **Polynomial division if improper:** If the degree of the numerator is greater than or equal to the degree of the denominator, preform polynomial long division on the numerator to obtain

$$\frac{Q(x)}{P(x)} = f(x) + \frac{Q_1(x)}{P(x)}$$

   Where $\deg(P(x))$ is less than $\deg(Q(x))$.

2. **Factor the denominator:** Factor the denominator into linear and quadratic factors of the following form

$$(px+q)^m \qquad \text{and} \qquad (ax^2+bx+c)^n$$

   where each factor is irreducible.

Any polynomial with real number coefficients can be completely factored into linear and quadratic factors.

3. **Linear factors:** For each linear factor of the form $(px+q)^m$, the partial fraction decomposition of this term must be

$$\frac{A_1}{(px+q)} + \frac{A_2}{(px+q)^2} + \cdots + \frac{A_m}{(px+q)^m}$$

4. **Quadratic factors:** For each factor of the form $(ax^2+bx+c)^n$, the partial decomposition of this term must be

$$\frac{B_1x+C_1}{(ax^2+bx+c)} + \frac{B_2x+C_2}{(ax^2+bx+c)^2} + \cdots + \frac{B_nx+C_n}{(ax^2+bx+c)^m}$$

In the steps above, the terms $A_1, ..., A_m$, $B_1, ..., B_n$, and $C_1, ..., C_n$ are unknown coefficients that must be solved for. The steps above give the general decomposition for the method of partial fractions, but we must also know how to solve for these coefficients. The following examples will show how to solve for these coefficients in a few different situations.

Example 1.12. *Evaluate* $\displaystyle\int \frac{5x^2+20x+6}{x^3+2x^2+x}\,dx$

Solution: First be must factor the denominator. Doing this we have

$$x^3+2x^2+x = x(x+1)^2$$

Therefore we have the following partial fraction decomposition

Notice that we must include a term for $(x+1)$ and $(x+1)^2$).

$$\frac{5x^2+20x+6}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

Now in order to solve for A, B, and C, we notice that if we get a common denominator on the right, we get a numerator of

$$A(x+1)^2 + Bx(x+1) + Cx$$

Therefore since the numerators of these two fractions must be the same, we have

$$5x^2+20x+6 = A(x+1)^2 + Bx(x+1) + Cx$$

Now notice that the key point in finding A, B, and C will be to choose convenient values of x to substitute into the equation above. If we substitute $x = 0$, then we have

$$6 = A$$

Now we will let $x = -1$. This gives

$$C = 9$$

Finally we will let $x = 1$ and use the fact that $A = 6$ and $C = 9$. This will give us the following

$$\begin{aligned} 31 &= 20 + 2B + 9 \\ B &= -1 \end{aligned}$$

Therefore coming back to our original problem, we have

$$\int \frac{5x^2+20x+6}{x^3+2x^2+x}\,dx$$

$$= \int \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}\,dx$$

$$= 6\ln|x| - \ln|x+1| + \frac{-9}{x+1} + C$$

$$= \boxed{\ln\left|\frac{x^6}{x+1}\right| + \frac{9}{x+1} + C}$$

During this next example we will see how to use partial fractions when dealing with repeated quadratic factor.

Example 1.13. *Evaluate* $\int \frac{8x^3+13x}{(x^2+2)^2}\,dx$.

Solution: Since the denominator is already factored and the degree of the numerator is less than the degree of the denominator, we have the following decomposition

$$\frac{8x^3+13x}{(x^2+2)^2} = \frac{Ax+B}{(x^2+2)} + \frac{Cx+D}{(x^2+2)^2}$$

For This example, we will start to solve for these coefficients by the same method as we did before. That is, we will set the numerators of the left and right hand side of the equation. Therefore we have

$$8x^3+13x = (Ax+B)(x^2+2)+Cx+D$$
$$8x^3+13x = Ax^3+2Ax+Bx^2+2B+Cx+D$$
$$8x^3+0x^2+13x+0 = Ax^3+Bx^2+(2A+C)x+(2B+D)$$

At this point, we will set the coefficients of the right and left hand sides to be equal. Therefore $B=0$ and $2B+D=0$. This gives us $D=0$ as well. Further, $A=8$ and $2A+C=13$, giving $C=-2$. Thus our original integral becomes

$$\int \frac{8x^3+13x}{(x^2+2)^2}\,dx$$

$$= \int \left(\frac{8x}{x^2+2} + \frac{-3}{(x^2+2)^2}\right) dx$$

$$= \boxed{4\ln(x^2+2) + \frac{3}{2(x^2+2)} + C}$$

Finally, we will show an example that seems to be a partial fractions problem, but as it turns out, partial fractions will not help.

Example 1.14. *Evaluate* $\int \frac{x+4}{x^2+2x+5}\,dx$.

Solution: The reason that partial fractions will not help with this problem is because the function above has already been decomposed. In this example, instead of trying to proceed by direct integration, we will "complete the square" in the denominator. Therefore we have

$$\int \frac{x+4}{(x+1)^2+4}\,dx$$

We have split the constant part of the numerator up so that we can use a u-substitution on the first integral.

$$\int \frac{x+1}{(x+1)^2+4}\,dx + \int \frac{3}{(x+1)^2+4}\,dx$$

Notice that we can use a u-substitution for which the left integral will become a natural logarithm and the right one will be an arctangent integral. Therefore we get

$$\int \frac{x+1}{(x+1)^2+4}\,dx + \int \frac{3}{(x+1)^2+4}\,dx$$

$$= \int \frac{u}{u^2+4}\,du + \int \frac{3}{u^2+4}\,du$$

$$= \frac{1}{2}\ln|u^2+9| - \frac{2}{3}\arctan u/3 + C$$

$$= \boxed{\frac{1}{2}\ln|(x+1)^2+9| - \frac{2}{3}\arctan((x+1)^2/3) + C}$$

Improper Integrals

The final topic in this chapter is the evaluation of improper integrals. The integrals that we plan to study will be split between **Type-I** improper integrals and **Type-II** improper integrals. To begin, let us define what we mean by the previous sentence.

Definition 1.15. *If $f(x)$ is continuous on the implied domain, then an improper integral of **Type-1** is an infinite integral the evaluation of one of the following*

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx$$

$$\int_{-\infty}^a f(x)\,dx = \lim_{t\to\infty}\int_t^b f(x)\,dx$$

$$\int_{-\infty}^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx + \lim_{t\to\infty}\int_{-t}^a f(x)\,dx$$

Definition 1.16. *An improper integral of **Type-II** is an integral given in one of the following forms*

1. *If f is continuous on $[a,b)$, but discontinuous at b, then the following integral is improper*

$$\int_a^b f(x)\,dx = \lim_{t\to b^-}\int_a^t f(x)\,dx$$

2. *If f is continuous on $(a, b]$, but discontinuous at a, then the following integral is improper*

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$$

3. *If f is continuous on $[a, c) \cup (c, b]$, but discontinuous at c, then the following integral is improper*

$$\int_a^b f(x)\,dx = \lim_{t\to c^-} \int_a^t f(x)\,dx + \lim_{t\to c^+} \int_t^b f(x)\,dx$$

In order to talk about evaluating these integrals, we must consider two more definitions. These are given below

Definition 1.17. *An improper integral of type I or II* ***converges*** *if the limits given in the definition exist and are finite.*

If the corresponding limits are infinite or do not exist, then the integral is said to ***diverge****.*

With this information, let us do some examples to illustrate how these concepts can play out.

Example 1.18. *Evaluate* $\displaystyle\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx$.

Solution: Notice that this function is continuous on the interval $(-\infty, \infty)$, therefore it is an improper integral of type I, and we can split the integral with $a = 0$, giving us

$$\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx$$

$$\lim_{t\to-\infty} \int_t^0 \frac{e^x}{1+e^{2x}}\,dx + \lim_{t\to\infty} \int_0^t \frac{e^x}{1+e^{2x}}\,dx$$

$$= \lim_{t\to-\infty} \Big[\arctan e^x\Big]_t^0 + \lim_{t\to\infty} \Big[\arctan e^x\Big]_0^t$$

$$= \lim_{t\to-\infty} \left(\frac{\pi}{4} - \arctan e^t\right) + \lim_{t\to\infty} \left(\arctan e^t - \frac{\pi}{4}\right)$$

$$= \frac{\pi}{4} - 0 + \frac{\pi}{2} - \frac{\pi}{4} = \boxed{\frac{\pi}{2}}$$

Since both of the integrals above are convergent, the original integral is convergent.

Example 1.19. *Evaluate* $\int_{-1}^{2} \frac{1}{x^3}\, dx$.

Note that this integral can be defined by using a Cauchy principle value. This term is named after Augustin Louis Cauchy, a French mathematician, who laid some of the foundation for the rigorous approach to calculus known as real analysis.

Solution: Notice that this integral is not a type I improper integral because the limits of integration are finite, however when $x = 0$ there is a discontinuity. This means that the integral is of type II. Therefore we must evaluate it as follows

$$\int_{-1}^{2} \frac{1}{x^3}\, dx$$

$$= \lim_{t\to 0^-} \int_{-1}^{t} \frac{1}{x^3}\, dx + \lim_{t\to 0^+} \int_{t}^{2} \frac{1}{x^3}$$

$$= \lim_{t\to 0^-} \left[-\frac{1}{2x^2} \right]_{-1}^{t} + \lim_{t\to 0^+} \left[-\frac{-1}{2x^2} \right]_{t}^{2}$$

$$= -\infty + \infty$$

Since neither of these integrals converge, the original integral is convergent.

Finally we will conclude this section with the comparison test and a result that will give you a set of test functions for the comparison test.

In logic, the reversal of the hypothesis and the conclusion of a statement is known as the converse. For more information on this, please see wikipedia

Theorem 1.20. *Suppose that f and g are continuous functions on $[a, \infty)$ with $f(x) \geq g(x) \geq 0$ for all $x \geq a$. Then*

1. *If $\int_a^\infty f(x)\, dx$ is convergent, then $\int_a^\infty g(x)\, dx$ is convergent.*
2. *If $\int_a^\infty g(x)\, dx$ is divergent, then $\int_a^\infty f(x)\, dx$ is divergent.*

In the two pieces of this theorem, the part of the sentence that comes before "then" is the hypothesis of the statement and the part that comes after is the conclusion. It is very important to note that we cannot switch the hypothesis and conclusion of these two statements. For example, if we look at point 1, if the integral of $f(x)$ diverges, it does not tell us if the integral of $g(x)$ converges.

To finish this section, we have the following theorem.

Proposition 1.21.

$$\int_1^\infty \frac{1}{x^p}\, dx \text{ is convergent for } p > 1 \text{ and divergent for } p \leq 1.$$

Please try to work through the proof of the result above noticing the difference in the anti-derivative when $p = 1$.

Applications of Integration

In this section we will consider a few applications of integration. We start with the calculating arc length of a function.

Arc Length

To begin this section, we will give a definition and the equations to calculate the arc length of a function.

Arc length Approximation

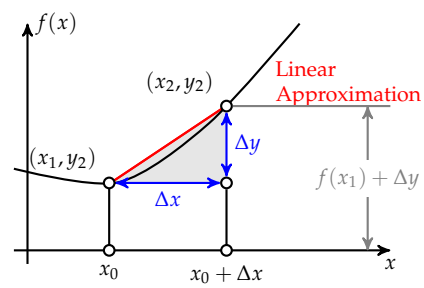
Definition 1.22. *A curve $f(x)$ in the x-y plane is called rectifiable if it has finite length.*

Theorem 1.23. *Let the function given by $y = f(x)$ represent a smooth curve on the interval $[a, b]$. The* ***arc length*** *of f between a and b is*

$$s = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$

Similarly, for a smooth curve given by $x = g(y)$, the arc length of g between c and d is

$$s = \int_c^d \sqrt{1 + [g'(x)]^2}\, dx$$



$$\Delta s = \sqrt{\Delta x^2 + \Delta y^2}$$

Proof. This will be a sketch of the proof of the above and should not be considered rigorous. Notice that we could approximate the arc length of a curve by performing a series of linear approximations as indicated to the right. If we break up the interval $[a, b]$ into n equally spaced intervals of length Δx, then our approximation would amount to the following sum

$$\sum_i^n \sqrt{\Delta x_i^2 + \Delta y_i^2}$$

where $\Delta y_i = f(x_i) - f(x_{i-1})$. Then notice that we can also rearrange this to be

$$\sum_i^n \sqrt{1 + \left[\frac{\Delta y_i}{\Delta x_i}\right]^2}\, \Delta x.$$

Now if we take the limit of this summation, letting n go to infinity, we get the integral

$$\int_a^b \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$$

□

Now we will work a few examples to illustrate how to use these equations.

Example 1.24. *Find the arc length of $f(x) = x^3/6 + 1/2x$ on the interval $[0.5, 2]$ Solution: To begin, we will find $f'(x)$. Differentiating this function, we have*

$$f'(x) = \frac{3x^2}{6} - \frac{1}{2x^2}$$

Therefore the arc length is calculated as follows

$$s = \int_{0.5}^{2} \sqrt{1 + \left[\frac{1}{2}\left(x^2 - \frac{1}{2}\right)\right]^2}\, dx$$

$$= \int_{0.5}^{2} \sqrt{\frac{1}{4}\left(x^4 + 2 + \frac{1}{x^4}\right)}\, dx$$

$$= \int_{0.5}^{2} \sqrt{\frac{1}{4x^4}\left(x^8 + 2x^4 + 1\right)}\, dx$$

$$= \int_{0.5}^{2} \frac{1}{2x^2}\sqrt{\left(x^4 + 1\right)^2}\, dx$$

$$= \int_{0.5}^{2} \frac{1}{2x^2}(x^4 + 1)\, dx$$

$$= \int_{0.5}^{2} \frac{1}{2}\left(x^2 + \frac{1}{x^2}\right) dx$$

$$= \frac{1}{2}\left(\frac{13}{6} - \frac{47}{24}\right)$$

$$= \frac{1}{2}\left[\frac{x^3}{3} - \frac{1}{x}\right]_{1/2}^{2}$$

$$= \boxed{\frac{33}{16}}$$

Notice that in this example, we pulled out the denominator in the square root term. This allowed us to factor of the polynomial

$$x^8 + 2x^4 + 1$$

as

$$(x^4 + 1)^2$$

Example 1.25. *Find the arc length of the graph* $(y-1)^3 = x^2$ *on the interval* $[0,8]$.

Solution: We will begin by solving for x *in terms of* y*. This gives*

$$x = \pm(y-1)^{3/2}$$

Since we are considering the graph of the function over $[0,8]$*, we will take the positive root of this. Now we will take the derivative of this function to get*

$$\frac{dx}{dy} = \frac{3}{2}\sqrt{y-1}$$

Since $y = 1$ *when* $x = 0$ *and* $y = 5$ *when* $x = 8$*, we will integrate* $\sqrt{1+(dx/dy^2)}$ *over the interval* $y \in [1,5]$*. Therefore*

$$\int_1^5 \sqrt{1+\left((3/2)^2(y-1)\right)}\,dx$$

$$= \int_1^5 \sqrt{(9/4)y-(5/4)}\,dx$$

$$= \frac{1}{2}\int_1^5 \sqrt{9y-5}\,dx$$

$$= \frac{1}{18}\left[\frac{(9y-5)^{3/2}}{3/2}\right]_1^5 dx$$

$$= \frac{1}{27}(40^{3/2}-4^{3/2}) \approx \boxed{9.073}$$

This will conclude the examples of this section. The equation for arc length is not difficult to remember, but it will take many examples to become proficient. Make sure to work through homework until you can work problems quickly.

Area of a Surface of Revolution

In this section we will give another application related to arc length. We will define the **surface of revolution** to be the surface resulting from revolving the graph of a function $f(x)$ around a line. The following gives a way to calculate the area of a surface of revolution.

Theorem 1.26. *Let* $y = f(x)$ *have a continuous derivative on the interval* $[a,b]$*. The area* S *of the surface of revolution formed by revolving the graph*

of f about a horizontal or vertical axis is given by

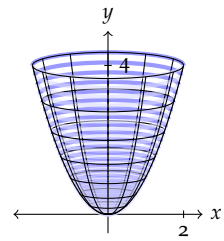
$$S = 2\pi \int_a^b r(x)\sqrt{1 + [f'(x)]^2}\, dx$$

where $r(x)$ is the distance between the graph of f and the axis of revolution. If $x = g(y)$ on the interval $[c, d]$, then the surface area is

$$S = 2\pi \int_c^d r(y)\sqrt{1 + [g'(x)]^2}\, dy$$

where $r(y)$ is the distance between the graph of g and the axis of revolution.

Example 1.27. *The arc of the parabola $y = x^2$ from $x = 0$ to $x = 2$ is rotated about the y-axis. Find the resulting surface area.*



Rotation of $f(x) = x^2$ about the y-axis.

Solution: To begin, notice that

$$\frac{dy}{dx} = 2x.$$

Therefore

$$S = \int_0^2 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$= 2\pi \int_0^2 x\sqrt{1 + 4x^2}\, dx$$

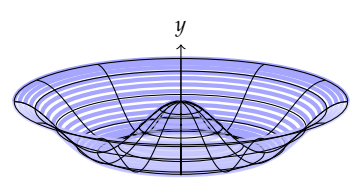
Now we will let $u = 1 + 4x^2$, and thus $du = 8x\, dx$. Then with the proper limits of integration, we have

$$S = \frac{\pi}{4} \int_1^{17} \sqrt{u}\, du$$

$$= \frac{\pi}{4}\left[(2/3)u^{3/2}\right]_1^{17}$$

$$= \boxed{\frac{\pi}{6}(17\sqrt{17} - 1)}$$

Example 1.28. *The arc of $y = \cos x$ from $x = 0$ to $x = 2\pi$ is rotated about the y-axis. Find the resulting surface area.*



Rotation of $f(x) = \cos x$ for $x \in [0, 2\pi]$ about the y-axis.

Solution: To begin, notice that

$$\frac{dy}{dx} = -\sin x$$

Therefore

$$S = \int_0^{2\pi} 2\pi x \sqrt{1 + \sin^2(x)}\, dx$$

For this example, we concede that the integral above cannot be evaluated analytically by our methods. However if we evaluate this integral numerically, we have the following

$$S \approx \boxed{150.82}$$

Example 1.29. *(**Gabriel's Horn**) For our final example, we will calculate the surface area of Gabriel's Horn. This is the surface area of revolution of $y = 1/x$ about the x-axis from $x = 1$ to $x = \infty$.*

Solution: To begin, notice that

$$\frac{dy}{dx} = \frac{-1}{x^2}$$

Therefore

$$S = \int_1^{\infty} 2\pi f(x) \sqrt{1 + f'(x)^2}\, dx$$

$$= 2\pi \int_1^{\infty} \left(\frac{1}{x}\right) \sqrt{1 + \frac{1}{x^4}}\, dx.$$

Now we will use the comparison test for integrals with the following observation

$$\sqrt{1 + \sqrt{1} x^4} > 1$$

on the interval $[1, \infty)$. Therefore we have

$$S = 2\pi \int_1^{\infty} \left(\frac{1}{x}\right) \sqrt{1 + \frac{1}{x^4}}\, dx$$

$$\geq 2\pi \int_1^{\infty} \frac{1}{x}\, dx = \boxed{\infty}$$

This concludes our examples in this section. Make sure to work through several examples to see the difficulties and intricacies of working surface area problems.

Application to Physics and Engineering

In this section we will consider two applications. The first will be to study fluid pressure and force. The section application will be to consider the center of mass of a uniform mass distribution.

Fluid Force and Fluid Pressure

In this section we will consider two principles that will help us to calculate pressure and force on an object submerged in a fluid. The first physical law that we will consider is **Pascal's Principle**. This states that **the pressure exerted by a fluid on an object at depth *d* is transmitted equally in all directions.**

The second principle that we need is that the fluid pressure increases the deeper that an object is submerged. This is seen in the following formula

Definition 1.30. *The **pressure** on an object at depth d is defined to be the force per unit are:*

$$P = \frac{F}{A} = \rho g d$$

where ρ is the density of the fluid and g is the acceleration constant due to gravity. The constant g is given by $g = 9.8m/s^2$ and the density of water is given by $\rho = 1000kg/m^3$. Likewise we can solve for pressure to give

$$PA = F = \frac{\rho g d}{A}.$$

Remark 1.31. *Notice that the principles above allow us to use integration to calculate the force exerted by a fluid on a vertical plate. This is given in the following definition.*

Definition 1.32. *The force F exerted by a fluid of density ρ on a vertical plane region from $y = a$ to $y = b$ is given by*

$$\rho g \int_a^b h(y) L(y)\, dy$$

where $h(y_0)$ gives the depth of the fluid at $y = y_0$, and $L(y_0)$ gives the horizontal length of the region at height $y = y_0$.

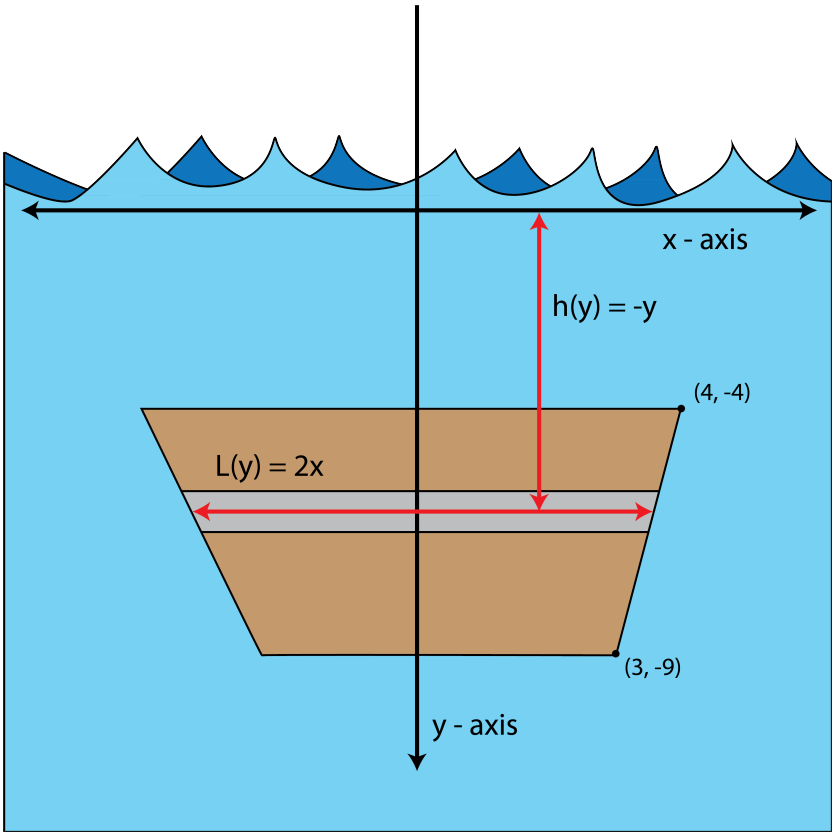
Now let us consider an example using this formula.

Example 1.33. *A trapezoidal plate having the dimensions $8m$ across on the top, $6m$ across on the bottom, and a height of $5m$, is submerged in a pool of water. Find the force on the plate if it is submerged vertically so that the top is $4m$ below the surface of the pool.*

Solution: First we must find formulas for $h(y)$ and $L(y)$. For $h(y)$, we will set the x-axis to sit on the top of the water, and we will center the y-axis to the center of the trapezoid. This will lead to

$$h(y) = -y.$$

Next we will notice that $L(x)$ is given by two times the x coordinate of an edge of the plate. This can be seen in the diagram on the following page.



Next, observe that we have $L(y)$ in terms of x. This means that we must find a relationship between the x coordinate of the right edge of the plate and y. To do this, we will use the point slope formula to find the equation for the line passing through the points $(3, -9)$ and $(4, -4)$. Therefore we have

$$y + 9 = 5(x - 3)$$

and this leads to the following expression for x

$$x = \frac{y + 24}{5}.$$

Therefore

$$L(y) = \frac{2}{5}(y + 24).$$

Finally, we must find the limits of integration. These are given by $a = -9$ and $b = -4$. Therefore we have the following calculation

$$F = \rho g \int_{-9}^{-4} h(y)L(y)\,dy$$

$$= 9800 \int_{-9}^{-4} (-y)\left(\frac{2}{5}\right)(y+24)\,dy$$

$$= 3920 \int_{-4}^{-9} y^2 + 24y\,dy$$

$$= 3920\left[\frac{y^3}{3} + 12y^2\right]_{-4}^{-9}$$

$$= 3920\left(\frac{1675}{3}\right)$$

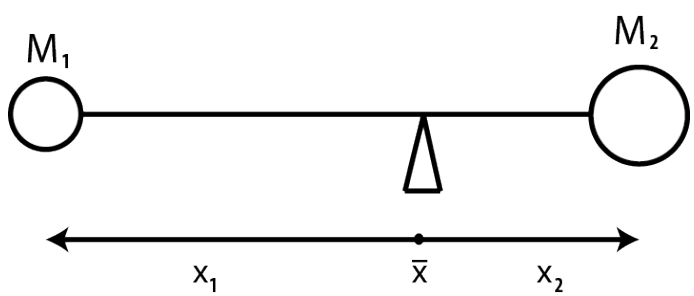
$$\approx \boxed{2.19 \times 10^6 N}$$

For this section, we have only completed one example, but please check moodle for other examples.

Moments and Centers of Mass

In this next section, we will consider the problem of finding the center of mass of a two dimensional plate, and the moment of a system of masses. We will begin this section by first considering a one dimensional problem.

To find the center of mass or balancing point of a one-dimensional system of two point masses of the following system



is given by the following formula

$$\overline{x} = \frac{M_1 x_1 + M_2 x_2}{M_1 + M_2}$$

This can be extended to several point masses $m_1, ..., m_n$ lying along the x-axis at positions $x_1, ..., x_n$. In this case, we have the following

$$\overline{x} = (1/M) \sum_{i=1}^{n} m_i x_i$$

Where $M = \sum_{i=1}^{n} m_i$ is the **total mass of the system** and the terms $m_i x_i$ are referred to as **moments** about the point $(0, 0)$.

This can be further extended to a system of masses in two dimensions. Consider a system of n particles with masses $m_1, ..., m_n$ located at the points $(x_1, y_1), ..., (x_n, y_n)$ in the xy-plane. Following the one-dimensional case, we define the **moment of the system about the y-axis** to be

$$M_y = \sum_{i=1}^{n} m_i x_i$$

and the **moment of the system about the x-axis** is

$$M_x = \sum_{i=1}^{n} m_i y_i.$$

Then the center of mass of this system is given by $(\overline{x}, \overline{y})$ where

$$\overline{x} = \frac{M_y}{m}, \qquad \overline{y} = \frac{M_x}{m},$$

and

$$m = \sum_{i=1}^{n} m_i$$

Centroid of a Lamina

Finally, we will extend our results to find the **centroid** of a flat plate or **lamina** in the xy-plane. The following gives us the result that we seek

Definition 1.34. *Let f and g be continuous functions such that $f(x) \geq g(x)$ on $[a, b]$, and consider the lamina of uniform density ρ bounded by the graphs of $y = f(x)$ and $y = g(x)$ and $a \leq x \leq b$.*

1. *The **moments about the x-axis and y-axis** are*

$$M_x = \rho \int_a^b \left[\frac{f(x) + g(x)}{2} \right] [f(x) - g(x)] \, dx$$

$$M_y = \rho \int_a^b x[f(x) - g(x)] \, dx.$$

2. *The* ***center of mass*** *is given by* $(\overline{x}, \overline{y})$*, where*

$$\overline{x} = \frac{M_y}{m},$$

$$\overline{y} = \frac{M_x}{m},$$

and

$$m = \rho \int_a^b [f(x) - g(x)]\, dx$$

These formulas can be simplified to the following

$$1) \quad \overline{x} = \frac{1}{A} \int_a^b x[f(x) - g(x)]\, dx$$

$$2) \quad \overline{y} = \frac{1}{A} \int_a^b \frac{1}{2} \left[f(x)^2 - g(x)^2 \right] dx$$

where A is the area between $f(x)$ and $g(x)$.

Now we will work through a few examples that show how such a calculation is performed.

Example 1.35. *Find the center of mass of the region bounded by the curves* $y = x^2$ *and* $x = y^2$*.*

Solution: To begin, we notice that the region is bounded between $x = 0$ *and* $x = 1$ *and* $\sqrt{x} \geq x^2$ *on this interval. Therefore we will let* $f(x) = \sqrt{x}$ *and* $g(x) = x^2$*. Now we will calculate the area of this region.*

$$A = \int_0^1 [\sqrt{x} - x^2]\, dx$$

$$= \left[\frac{2}{3} x^{3/2} - dfrac13x^3 \right]_0^1$$

$$= \frac{1}{3}$$

Now we will calculate $\overline{x}$ and $\overline{y}$.

$$\overline{x} = 3\int_0^1 x[\sqrt{x} - x^2]\,dx$$

$$= 3\int_0^1 x^{3/2} - x^3\,dx$$

$$= 3\left[\left(\frac{2}{5}\right)x^{2/5} - \left(\frac{1}{4}\right)x^4\right]_0^1$$

$$= 3\left(\frac{3}{20}\right) = 9/20$$

$$\overline{y} = \frac{3}{2}\int_0^1 [x - x^4]\,dx$$

$$= \frac{3}{2}\left[\left(\frac{1}{2}\right)x^2 - \left(\frac{1}{5}\right)x^5\right]_0^1$$

$$= \frac{3}{2}\left(\frac{3}{10}\right) = \frac{9}{20}$$

Therefore we have

$$(\overline{x}, \overline{y}) = \boxed{\left(\frac{3}{20}, \frac{3}{20}\right)}$$

Example 1.36. *Find the centroid of the region bounded by $y = \sin x$ and $y = \cos x$ between $x = 0$ and $x = \pi/4$.*
Solution: To begin, we calculate the area between these two curves. Since $\cos x \geq \sin x$ on this interval, we have

$$A = \int_0^{\pi/4} [\cos x - \sin x]\,dx$$

$$= \Big[\sin x + \cos x\Big]_0^{\pi/4}$$

$$= \sqrt{2} - 1$$

Now we will calculate $\overline{x}$ and $\overline{y}$. Notice that in the following calculations,

we will use integration by parts.

$$\overline{x} = \frac{1}{\sqrt{2}-1}\int_0^{\pi/4}[x(\cos x - \sin x)]\,dx$$

$$= \frac{1}{\sqrt{2}-1}\left[x\big(\sin(x)+\cos(x)\big)\Big|_0^{\pi/4} - \int_0^{\pi/4}(\cos x + \sin x)\,dx\right]$$

$$= \frac{1}{\sqrt{2}-1}\left[(\pi/4)\left(\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}\right) - 0\cdot\left(0+1\right) + \big[-\sin(x)+\cos(x)\big]_0^{\pi/4}\right]$$

$$= \frac{1}{\sqrt{2}-1}\left[(\pi/4)\left(-\sqrt{2}\right) + \left(-\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}-0-1\right)\right]$$

$$= \frac{\pi\sqrt{2}-4}{4\sqrt{2}-4}$$

and

$$\overline{y} = \frac{1}{\sqrt{2}-1}\int_0^{\pi/4}\frac{1}{2}\left[\cos^2 x - \sin^2 x\right]dx$$

$$= \frac{1}{\sqrt{2}-1}\int_0^{\pi/4}\frac{1}{4}(1+\cos(2x)-1+\cos(2x))\,dx$$

$$= \frac{1}{\sqrt{2}-1}\int_0^{\pi/4}\frac{1}{2}\cos(2x)\,dx$$

$$= \frac{1}{2(\sqrt{2}-1)}\big(\sin x\cos x\big)\Big|_0^{\pi/4}$$

$$= \frac{1}{4(\sqrt{2}-1)}$$

Therefore the centroid of this region is

$$(\overline{x},\overline{y}) = \boxed{\left(\frac{\pi\sqrt{2}-4}{4\sqrt{2}-4}, \frac{1}{4(\sqrt{2}-1)}\right)}$$

This concludes our excursion into applications of calculus to physics. Make sure to work through several example in order to gain proficiency in these exercises.

Differential Equations

Introduction

In this chapter, we will discuss an important area of mathematics known as differential equations. Many subjects outside of math use models to predict future outcomes. Differential equations are particularly important in modeling quantities that change with time. The following are important differential equations that are used to model phenomena in various fields.

- Schrödinger's equation (physics)
- Navier-Stokes equation (physics)
- Rate equation (chemistry)
- Differential form of Gibbs Equation (chemistry)
- Solow-Swan Model (economics)
- Black-Scholes (economics)
- Hodgkin-Huxley model (biology)
- Lotka-Volterra (biology)

Let us now consider the vocabulary of this chapter. A **differential equation** is an equation that contains a **dependent variable**, usually "y", its derivatives y', y'',..., $y^{(n)}$, and possibly an **independent variable** x. The following are examples of differential equations

$$y'' - y = 1 \qquad\qquad y' + y = \sin x$$

$$y'' - y = 4e^{-}x \qquad\qquad y' + e^{x}y = \arctan(x)$$

The **order** of a differential equation is the order of the highest derivative in the differential equation. The equations in the left column are second order differential equations and the right column contains first order differential equations.

A **solution to a differential equation** is a function f such that the equation is satisfied when y is replaced by $f(x)$. For example $f_1(x) = \sin(x)$ is a solution to the equation $y'' + y = 0$ since

$$f_1''(x) = -\sin(x).$$

Notice that in this example, $f_2(x) = \cos x$ is another solution to this differential equation. It turns out that $Af_1(x)$ and $Bf_2(x)$ are also solutions, where A and B are constants. The following example gives a way to show whether or not a function is a solution to a differential equation.

Example 1.37. *Show that $y_g = A\sin x + B\cos x$ is a solution to the differential equation $y'' + y = 0$.*

Solution: Since $y_g = A\sin x + B\cos x$ and $y_g'' = -A\sin x - B\cos x$, we have

$$y_g'' + y_g = (-A\sin x - B\cos x) + (A\sin x + B\cos x) = 0.$$

Since y_g satisfies the equation $y'' + y = 0$, it follows that y_g is a solution to the differential equation.

In the example given above, the set $\mathcal{F}_1 = \{A\sin x : A \text{ is a constant}\}$ is called a **family of solutions**. Similarly $\mathcal{F}_2\{B\cos x : B \text{ is a constant}\}$ is also a family of solutions. It turns out that the differential equation $y'' + y = 0$ only has two families of solutions and combinations of functions from both of these families is also a solution. Now we will introduce a definition that will allow us to discuss solutions.

Definition 1.38. *If $\mathcal{F}_1$ and $\mathcal{F}_2$ are two sets of functions, then a **linear combination** of these sets is a function of the form*

$$h(x) = a \cdot f(x) + b \cdot g(x)$$

Where a and b are constants.

Example 1.39. *Using the example above,*

$$h(x) = 2\sin x + 3\cos x$$

is a linear combination of $\sin x$ *and* $\cos x$.

Finally we can restate the result of example 1.37 succinctly as *any linear combination of* $\sin x$ *and* $\cos x$ *is a solution of* $y'' + y = 0$.

This leads to a natural question. If $\mathcal{F}_1$ and $\mathcal{F}_2$ are two families of solutions to a differential equation, is it true that any linear combination

from these two families is also a solution? In general, the answer to this is question is no. If however, the differential equation is "linear" (to be defined later), then the answer is yes. Now let us consider another definition

Definition 1.40. *If $\mathcal{F}_1$,..., $\mathcal{F}_n$ are the families of solutions to a linear differential equation, then the* ***general solution*** *is defined to be*

$$a_1 f_1(x) + a_2 f_2(x) + ... + a_n f_n(x)$$

Where the a_i's are constants and $f_i \in \mathcal{F}_i$.

With all this talk about general solutions and family of solutions, one might expect that differential equations have an infinite number of solutions. This is true for a general differential equation, but often there is additional information that restricts us to have only one solution. The piece of information that gives us a **particular solution** is called an **initial condition**. An initial value is often given by one of the following conditions

$$y(x_0) = y_0 \qquad \text{or} \qquad y(t_0) = y_0$$

An initial value along with a differential equation is called an **initial value problem**. We will now give two examples of how to solve an initial value problem.

Example 1.41. *For the differential equation $xy' - 3y = 0$ and initial value $y(-3) = 2$, verify that $y_1 = Cx^3$ is a solution. Further, use the initial value find the particular solution.*

Solution: Notice that $y_1' = 3Cx^2$, therefore

$$xy_1' - 3y_1 = 3Cx^3 - 3Cx^3 = 0$$

and this shows that y_1 a solution to $xy' - 3y = 0$. Now we may apply the initial value to obtain the following

$$\begin{aligned} 2 &= y_1(-3) \\ 2 &= C(-3^3) \\ -\frac{2}{27} &= C \end{aligned}$$

Therefore the ***particular solution*** *is*

$$\boxed{y = -\frac{2}{27}x^3}$$

Example 1.42. *Find the particular solution to the differential equation $y'' + y = 0$ with the initial conditions $y(0) = 1$ and $y''(\pi/4) = 2$*

Solution: From a previous example, we know that the general solution to this differential equation is $y_1 = a \sin x + b \cos x$. Now we can apply the initial value $y(0) = 1$ to obtain

$$\begin{aligned} A\sin(0) + B\cos(0) &= 1 \\ B &= 1 \end{aligned}$$

Now we can apply the initial value $y(\pi/4) = 2$ and use the fact that $B = 1$ to obtain

$$\begin{aligned} A\sin(\pi/4) + \cos(\pi/4) &= 2 \\ A(\sqrt{2}/2) + \sqrt{2}/2 &= 2 \\ A + 1 &= 4/\sqrt{2} \\ A &= 2\sqrt{2} - 1 \end{aligned}$$

Therefore the particular solution is

$$\boxed{y = (2\sqrt{2} - 1)\sin x + \cos x}$$

We will conclude this section by conceding the fact it can be very difficult to solve certain differential equations, and in some cases, it can be impossible. In the next section we will find a numerical method for solving first order differential equations.

Slope Fields and Euler's Method

Separable Equations

Models for Population Growth

For this section, we will be brief. We will introduce the two models of population growth that are of interest to us and then we will work and example of each. To begin, let us define the two models.

Definition 1.43. *The initial-value problem*

$$\frac{dP}{dt} = kP \qquad P(0) = P_0$$

*is the differential equation that models exponential growth/decay. k is known as the **relative growth rate**. If $k > 0$, this represent **exponential growth**. If $k < 0$, it represents **exponential decay**.*

Definition 1.44. *The initial-value problem*

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right) \qquad P(0) = P_0$$

is the differential equation that models ***logistic growth****.*

The solution to the exponential growth/decay problem is found with a simple separation of variables. The solution is

$$\boxed{P(t) = P_0 e^{kt}}$$

The solution to the logistic growth problem is found using a separation of variables along with partial fraction decomposition. The solution is

$$\boxed{P(t) = \frac{M}{1 + Ae^{-kt}} \qquad \text{where} \qquad A = \frac{M - P_0}{P_0}}$$

To begin, let us do an example of exponential decay.

Example 1.45. *(**Newton's Law of Cooling**) Let T represent the temperature of an object in a room whose temperature is kept at a constant 60^oF. Newton's law states that the rate of change in the temperature of an object is proportional to the difference between the temperature of the object and that of the surrounding medium. If the object cools from 100^oF to 90^oF in 10 minutes, how much longer will it take for the temperature to decrease to 80^oF? Solution: To begin, notice that our differential equation is*

$$\frac{dT}{dt} = k(T - 60), \qquad 80 \leq T \leq 100$$

We will now solve this initial value problem with separation of variables.

$$\frac{dT}{dt} = k(T - 60)$$

$$\frac{dT}{T - 60} = k\,dt$$

$$\int \frac{1}{T - 60}\,dT = \int k\,dt$$

$$\ln|T - 60| = kt + C_1$$

Since $T > 0$, $|T - 60| = T - 60$, and we have the following

$$T - 60 = e^{kt + C_1} \implies T = 60 + Ce^{kt}$$

From the initial condition $T(0) = 100$, *we have* $100 = 60 + Ce^0$. *Thus* $C = 40$. *Further, since* $T(10) = 90$, *we have*

$$\begin{aligned} 90 &= 60 + e^{10k} \\ 30 &= 40e^{10k} \\ k &= (1/10)\ln 3/4 \end{aligned}$$

Therefore we have the following solution

$$\begin{aligned} 80 &= 60 + 40e^{\ln((3/4)^{1/10})t} \\ 20 &= 40e^{\ln((3/4)^{1/10})t} \\ 1/2 &= e^{\ln((3/4)^{1/10})t} \\ \ln(1/2) &= \ln((3/4)^{1/10})t \\ t = \frac{\ln(1/2)}{\ln((3/4)^{1/10})} &\approx \boxed{24.09 \textit{ minutes}} \end{aligned}$$

This shows that it will take 24.09 minutes for the object to reach $80^o F$.

Example 1.46. *The Pacific halibut fishery has been modeled by the differential equation*

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{M}\right)$$

where $y(t)$ *is the biomass (total mass of the members of the population) in kilograms at time* t *(measured in years), and* $k = 0.71$

(a) If $y(0) = 2 \times 10^7$ *kg, find the biomass a year later.*

(b) How long will it take for the biomass to reach 4×10^7 *kg?*

Solution: We will solve the problem using separation of variables along with partial fraction decomposition. To begin, we have

$$\frac{dy}{dt} = \left(\frac{k}{M}\right) \cdot y(M-y)$$

$$\frac{M\ dy}{y(M-y)} = k\,dt$$

$$\int \frac{M}{y(M-y)}\,dy = \int k\,dt$$

$$\int \left[\frac{1}{y} + \frac{1}{M-y}\right] dy = kt + C_1$$

$$\ln|y| - \ln|M-y| = kt + C_1$$

$$\ln\left|\frac{M-y}{y}\right| = -kt + C_1$$

$$\frac{M-y}{y} = e^{-kt+C_1} = Ce^{-kt}$$

$$M - y = Cye^{-kt}$$

$$(Ce^{-kt}+1)y = M$$

$$y = \frac{M}{1+Ce^{-kt}}$$

Now we will use the initial value $y(0) = 2.7 \times 10^7$ *and* $M = 8 \times 10^7$ *to obtain*

$$C = \frac{8 \times 10^7 - 2 \times 10^7}{2 \times 10^7} = 3$$

Therefore after one year, the biomass will be

$$y = 8 \times 10^7/(1+3e^{-0.71}) \approx \boxed{3.23 \times 10^7 \text{ kg}}$$

Linear Equations

A **first-order linear differential equation** is an equation of the following form

$$y' + P(x)y = Q(x)$$

where P and Q are continuous function on a given interval. Linear first order equations show up in many applications to science. To solve such a differential equation, we will use the method of multiplying the differential equation by an **integrating factor**. To see how this will be achieved, consider the following example

Example 1.47. *Using the product rule and the fundamental theorem of calculus, we have*

$$\frac{d}{dx}\left[ye^{\int P(x)\,dx}\right] = y'e^{\int P(x)\,dx} + y\frac{d}{dx}(e^{\int P(x)\,dx}$$

$$= y'e^{\int P(x)\,dx} + ye^{\int P(x)\,dx}\left[\frac{d}{dx}\int P(x)\,dx\right]$$

$$= e^{\int P(x)\,dx}\left[y' + P(x)y\right]$$

Therefore we have

$$e^{\int P(x)\,dx}\left[y' + P(x)y = Q(x)\right]$$

$$\iff \frac{d}{dx}\left[ye^{\int P(x)\,dx}\right] = e^{\int P(x)\,dx}Q(x)$$

$$\implies \frac{d}{dx}\left[ye^{\int P(x)\,dx}\right] = e^{\int P(x)\,dx}Q(x)ye^{\int P(x)\,dx} = \int e^{\int P(x)\,dx}Q(x)\,dx$$

This leads to the following definition

Definition 1.48. *The **integrating factor** of a first order linear differential equation is given by*

$$I(x) = e^{\int P(x)\,dx}$$

To solve a linear first order differential equation, we must multiply both sides by $I(x)$ and integrate.

Predator-Prey Systems

Glossary of Terms

- **Autonomous Equation** - An autonomous equation is a differential equation that depends only upon the dependent variable. In

general, a first order autonomous equation is of the form

$$y' = f(y)$$

- **Carrying Capacity** - The carrying capacity of the logistic growth model is the maximum population that the environment is capable of sustaining over a long period of time.

- **Ordinary Differential Equation** - A differential equation is a mathematical equation involving a dependent variable, its derivatives, and possibly an independent variable. An ordinary differential equation does not involve the derivatives of an independent variable.

- **Direction Field** - A graphical representation of the solutions of a first order differential equation. It is created by placing small line segments, that represent slope, on several points in the xy-plane.

- **Equilibrium Solution** - An equilibrium solution is a solution that is constant with respect to the independent variable of a differential equation.

- **Euler's Method** - Euler's method is a numerical method for solving first order differential equations. (Pronounced as "Oil-ers" not "Yew-lers")

- **Explicit Solution** - A solution that depends only upon the independent variable.

- **Exponential Growth** - A model for population growth that assumes that the change in population is proportional to a growth constant times the current population.

- **Family of Solutions** - A set of solutions to a differential equation that differ by multiplication or addition of a constant. It may turn out that a differential equation has more than one family of solutions.

- **General Solution** - A general solution to a differential equation is of the form

  $$y_1 + y_2 + ... + y_n$$

  where each y_i are from a family of solutions.

- **Implicit Solution** - A solution that depends on both the independent and dependent variable of the differential equation.

- **Initial Condition** - An initial condition is the value of the independent variable at a particular value of the dependent variable. This is often given in the following form

$$y(t_0) = y_0 \qquad \text{or} \qquad y(x_0) = y_0$$

- **Initial Value Problem** - An initial value problem is a differential equation along with an initial value. An initial value is needed to produce a unique solution to a differential equation.

- **Integrating factor** - A function that is chosen to aid in solving a linear differential equation. For a linear differential equation of the form

$$y' + P(x)y = Q(x)$$

  the integrating factor is

$$I(x) = e^{\int P(x)\,dx}$$

- **Linear First Order Equation** - A differential equation of the form

$$y' + P(x)y = Q(x)$$

  where $P(x)$ and $Q(x)$ are continuous function depending only upon x.

- **Logistic Model** - A model for population growth that assumes exponential growth with the exception that population must be bounded by a constant known as the carrying capacity.

- **Lotka-Volterra Equations** - A system of two equations that describe the relationship between two populations. One being a predator population, and the other being the prey population.

- **Numerical Solution** - A method/algorithm designed to approximate the solution of a differential equation. For example see Euler's method.

- **Order of a Differential Equation** - The order of a differential equation is the highest derivative in the equation.

- **Orthogonal Trajectory** - An orthogonal trajectory of a family of curves is a curve that intersects each curve of the family of curves at a ninety degree angle.

- **Particular Solution** - A particular solution is a solution curve that is determined by an initial condition.

- **Phase Plane** - Given a system of differential equations with dependent variables R and W, the RW-plane is called

- **Phase Portrait** - A geometric representation of the trajectories of a system of differential equations. If the dependent variables of the system are R and W, then this consists of equilibrium points, the slope field determined by dW/dR, and typical trajectories in the phase plane.
- **Phase Trajectory** - A solution curve drawn in the phase plane is called a phase trajectory.
- **Predator-Prey Equations** - See Lotka Volterra Equations
- **Relative Growth Rate** - The constant given in the exponential growth model.
- **Slope Field** - See direction field.